



CHECKLIST

SMALL CRAFT - SEACOCKS AND THROUGH-HULL FITTINGS

Ref.: EN ISO 9093:2021 (ISO 9093:2020)

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
Watercraft model name:	



	Subject to check	Clause	Requirements	Checked ?
1	Through-hull fittings, seacocks, hose connections and fittings below the maximum load waterline are corrosion-resitant.	4.1	[Yes / NA ?]	
2	Through-hull fittings, seacocks, hose connections and fittings above the maximum load waterline are corrosion-resitant or are protected against corrosion.	4.1	[Yes / NA ?]	
3	In case of material combinations the possibility of galvanic interaction has been taken into account. If necessary they are galvanically isolated.	4.2	[Yes / NA ?]	
4	If swelling and/or seizure of material combinations is possible it is taken into account. The material combination does not prevent any device or system from functioning.	4.2	[Yes / NA ?]	
5	If aluminum is used in the system, non-ferros metals are not used.	4.2	[Yes / NA ?]	
6	Any through-hull fitting cannot be dislodged by outside forces due to operation of the fitting and components attached.	5.2	[Yes / NA ?]	
7	The design of the seacock allows operation under any condition to be encountered under normal service condition.	6.1	[Yes / NA ?]	
8	The design of the seacock allows a visual check of the open and closed position for the seacock.	6.1	[Yes / NA ?]	
9	Easy operation of the seacock is given by the length of the handle and clear distance between handle and attached hose.	6.1	[Yes / NA ?]	
10	The seacock is designed so that no part can come loose under normal service conditions.	6.1	[Yes / NA ?]	
11	Clamping lenght: - not less 25 mm for hoses of less than 25,4 mm (1 in); - not less 35 mm for hoses above 25,4 mm (1 in).	7.1	[Yes / NA ?]	
12	Where the centerline of the through-hull fitting is lower than the heeled waterline: Hoses of more than 25,4mm are fastened through double clamps or a single T bolt clamp.	7.2.1	[Yes / NA ?]	
13	Where the centerline of the through-hull fitting is lower than the heeled waterline: Hoses of 25,4 mm and below are fastened through double clamps.	7.2.1	[Yes / NA ?]	
14	Plain pipe ends are always secured with double clamps.	7.2.2	[Yes / NA ?]	
15	Clamps depending on spring tension are not used.	7.2.4	[Yes ?]	
16	Where water scoops and outboard strainers are fitted and the intake pipe and through-hull connection cannot be cleaned, they shall be removable.	8	[Yes / NA ?]	

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17	Where the fitting of a seacock or through-hull fitting impairs the required strength of the hull, a local reinforcement compensates for the loss of strength.	9.1	[Yes / NA ?]	
18	For hulls that are build in sandwich construction, the area around the fitting needs to be made as single-shell construction with local reinforcement, or an appropriate core material is used.	9.1	[Yes / NA ?]	
19	Through-hull fittings and seacocks are watertight and installed as to prevent loosening under normal operating conditions.	9.2.1	[Yes ?]	
20	Seacocks are accessible for inspection.	9.2.2	[Yes / NA ?]	
21	Seacocks below the maximum load waterline are readily accessible for inspection and operation.	9.2.3	[Yes / NA ?]	
22	Any seacock is securely fastened to the hull so that easy operation is possible without destroying the hull structure, the seacock itself or water integrity.	9.2.4	[Yes / NA ?]	
23	Seacocks and through-hull fittings are located so as to minimise the likelihood of damage to them or inadvertent operation.	9.2.6	[Yes ?]	

The following questions shall be filled in by the watercraft manufacturer and appropriate documentation shall be submitted to the inspector for verification.

	Subject to check	Clause	Requirements	Checked ?
24	Metallic components are corrosion resistant and tested as per Annex B of ISO 9093 with approval.	4.3	[Yes / NA ?]	
25	Non-metallic parts are stabilized against oxidation and UV and tested as per Annex C of ISO 9093 with approval.	4.3	[Yes / NA ?]	
26	Seacocks and through-hull fittings fulfil the requirements of the strength test. Proof by the supplier has been submitted.	4.4	[Yes ?]	
27	The range of operating temperatures has been verified for seacocks and through-hull fittings. Proof of compliance has been submitted.	4.5	[Yes ?]	
28	The detailed design requirements of the trough-hull fitting for the stem, flange diameter and finish are met.	5.3	[Yes ?]	
29	The seacock assembly has a minimum threaded length for attachment to through-hull fittings.	6.2	[Yes ?]	
30	Hose fitting ends are either ribbed, serrated or if nominal size 31,75 mm (1 1/4 in) and above, plain pipe end.	7.1	[Yes ?]	
31	Metallic hose clamps are made entirely of stainless steel, type Cr18 Ni8 or other material with equal or higher strength and corrosion resistance, and they are reusable.	7.2.3	[Yes ?]	
32	The boat manufacturer has tested the assembly as installed as per Annex A of ISO 9093 and a test report has been submitted.	9.2.5	[Yes ?]	

Instructions/Warnings to be included in the owner's manual

33	Location of the seacocks.	11	[Yes ?]	
34	Operating instructions, if relevant.	11	[Yes / NA ?]	
35	Maintanance and servicing instructions.	11	[Yes ?]	
36	Annual visual inspections.	11	[Yes ?]	
37	Physical test of seacocks functionallity.	11	[Yes ?]	
38	Warning notice as to keeping seacocks closed when not needed for operation to minimize the risk of flooding.	11	[Yes ?]	